Kevin H. Nguyen

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EDUCATION

Georgia Institute of Technology **Atlanta, GA**

B.S. in Mechanical Engineering | GPA: 3.29 *Expected May 2028*

- **Relevant Coursework:** Statics, Eng Graphics & Design, Materials Science, Circuits & Electronics, Computing for Engineers

EXPERIENCE

Georgia Tech — Microelectronics and Emerging Technologies Thermal Laboratory **Atlanta, GA**

Thermal/Fluids Research Intern *May 2026 – Present*

- Designed 5 liquid cold-plate geometries to dissipate chip heat by optimizing fin layouts across 25 ANSYS Fluent conjugate heat transfer simulations, cutting simulated peak chip temperature 3.5°C, thermal resistance 8%, and pressure drop 15% vs. baseline
- Modeled the 5 geometries in SolidWorks for additive manufacturing as part of an ASME/IEEE ITherm 2027 submission
- Validated CFD accuracy with mesh-refinement and energy-balance checks, limiting fine-mesh temperature change to 0.6°C

HyTech Racing — Formula SAE **Atlanta, GA**

Suspension Engineer *August 2025 – Present*

- Redesigned rear suspension bay in CATIA, relocating the differential ~40 mm forward and widening the chassis to package a 5:1 final-drive sprocket, eliminating ~2.5 kg of planetary-reduction hardware and enabling a higher rear roll center.
- Simulated roll, heave, and lateral load transfer in MATLAB to raise rear roll-center height from ~35 to 52 mm, shifting rear load-transfer distribution by ~4% to tune vehicle balance toward the team's oversteer target.
- Moved dampers forward to raise torsional rigidity and lower CoG, cutting ground clearance from ~50 to 27 mm (~46%)

Georgia Tech Undergraduate Research **Atlanta, GA**

Mechanical Design Researcher, Ground Breaking Infrastructure Program *August 2025 – May 2026*

- Developed a free-fall penetrometer by integrating a 50 kg load cell and IMU with an ESP32 data acquisition system that logs over Bluetooth with <7% average deviation, to gather ground properties across soil and sand in hard-to-access environments
- Designed a modular four-section SolidWorks enclosure by iterating three 3D-printed revisions across multiple materials, selecting a durable resin to accommodate the sensor electronics packaging and teammates' design requirements
- Developed MATLAB workflows to process 4.5 kHz IMU data from 100+ soil and sand penetration tests (2-6 in. depth) and, with a three-person research team, a low-cost architecture that cut projected system cost by more than $400

PROJECTS

Autonomous Robot Design Competition **Atlanta, GA**

Mechanical Design & Mechatronics Team Member *May 2026*

- Designed a robot by down-selecting 3 concepts per subsystem (drivetrain, actuation, manipulation) via a House of Quality (30+ requirements) and iterating ~3 SolidWorks revisions each within a 12 × 18 × 24 in envelope to complete the competition tasks
- Fabricated 10+ custom components via 3D printing, laser cutting, and mill/lathe machining, reducing task cycle time 33%
- Programmed Arduino C++ stop-sensor logic and 4-level lidar height detection, increasing autonomous success from 30% to 80%

Smart RGB Laptop Cooling Dock **Atlanta, GA**

Personal Project *August 2026*

- Designed and fabricated a modular 3D-printed cooling dock for 16-inch laptops in SolidWorks, integrating dual 120 mm fans, an OLED display, and removable electronics mounts to cut CPU/GPU-zone temperature 6°C (35.9°C to ~30°C) under heavy load
- Programmed STM32 temperature monitoring and 3-mode (manual, fixed, closed-loop) PWM fan control, tuned in MATLAB

Assistive Stair-Climbing Device (Conceptual CAD Project) **Atlanta, GA**

Mechanical Design Team Member *August 2025 – December 2025*

- Designed 5+ of the 24+ parts of a conceptual 3-subsystem tracked stair-climbing assistive device in SolidWorks as a 3-person team, following the full engineering design process from ideation through iteration, technical drawings, and animation

LEADERSHIP

Georgia Tech Vietnamese Student Association **Atlanta, GA**

Sponsorship Co-Director *August 2025 – Present*

- Secured ~$15,000 in funding for the annual Moon Festival by coordinating with local businesses, sponsors, and a 10-member executive board, supporting an event with ~4,000 attendees and 30 external organizations

TECHNICAL SKILLS

- **CAD & Mechanical Design:** SolidWorks, CATIA, Engineering Drawings, GD&T, Mechanical Packaging, DFA
- **Simulation & Thermal:** ANSYS Fluent (CFD), FEA, Conjugate Heat Transfer, Liquid Cooling, Thermal Testing
- **Analysis & Testing:** MATLAB, Python, Excel, Data Analysis, Suspension Kinematics, Prototype Testing & Validation
- **Mechatronics:** STM32, ESP32, Arduino C/C++, Sensor Integration, Data Acquisition, Closed-Loop/PWM Control
- **Fabrication:** FDM 3D Printing, Laser Cutting, Mill/Lathe Machining, Soldering, Design for Additive Manufacturing